
Knowledge, skills and understanding

Content for both components

Introduction

This qualification can be followed through the broad-based title of Art, Craft and Design or through one of five endorsed titles: Fine Art, Graphic Communication, Textile Design, Three-dimensional Design and Photography. Details of all titles are set out in the following pages. When students are entered for an endorsed title, they will follow content in that endorsed title for both components. Students undertaking the Art, Craft and Design title will work in disciplines chosen from **at least two** of the other five endorsed titles for Component 1. For Component 2, students undertaking the Art, Craft and Design title may choose to produce work in **one or more** disciplines, giving the option to specialise if desired. For the remaining endorsed titles, students are expected to cover similar content for both components.

Art and Design titles

All titles explore practical and critical/contextual work through a range of processes and media.

Art, Craft and Design: through a broad-based course of study, developed through a range of 2D, 3D and time-based processes and media, drawn from disciplines from **at least two** of the other five endorsed titles in Component 1, with study of **one or more** disciplines in Component 2.

Fine Art: through the disciplines of painting and drawing, printmaking, sculpture, lens-based image making.

Graphic Communication: through the disciplines of advertising, illustration, branding, information design.

Textile Design: through the disciplines of textiles for interiors, fine art textiles, fashion textiles.

Three-dimensional Design: through the disciplines of spatial design, product design, design crafts.

Photography: through the disciplines of film-based photography, digital photography, film and video.

For all titles, students are required to develop practical and theoretical knowledge and understanding of:

- relevant materials, processes, technologies and resources

- continuity and change in different genres, styles and traditions
- a working vocabulary and specialist terminology.

For all titles, students are required to develop the skills to:

- record experiences and observations in a variety of ways using drawing or other appropriate visual forms; undertake research; gather, select and organise visual and other appropriate information
- explore relevant resources; analyse, discuss and evaluate images, objects and artefacts; make and record independent judgements
- use knowledge and understanding of the work of others to develop and extend thinking and inform own work
- generate and explore potential lines of enquiry using appropriate media and techniques
- apply knowledge and understanding in making images and artefacts; review and modify work; plan and develop ideas in the light of their own and others' evaluations
- organise, select and communicate ideas, solutions and responses, and present them in a range of visual, tactile and/or sensory forms.

Developing students' knowledge, skills and understanding in art and design

Students need opportunities to generate ideas and research from primary and contextual sources, record their findings, experiment with media and processes, and develop and refine their ideas towards producing outcome(s). It is essential that students review their progress at appropriate points in the development of their work.

Each component aims to develop students' ability to generate and develop ideas for their practical work and to build contextual understanding, from either a self-selected or teacher-negotiated focus. Students will be required to:

- develop ideas through sustained and focused investigations informed by contextual and other sources, demonstrating analytical and critical understanding
- explore and select appropriate resources, media, materials, techniques and processes, reviewing and refining ideas as work develops
- record ideas, observations and insights relevant to intentions, reflecting critically on work and progress
- present a personal and meaningful response that realises intentions and, where appropriate, makes connections between visual and other elements .

For all titles, students should also:

- understand how ideas generate starting points for art and design practice and form an integral part of the creative process
- understand how images and artefacts relate to ethical and political contexts

- develop their visual language to communicate ideas
- demonstrate skilful use of the formal elements, including line, tone, colour, shape, pattern, texture, form and structure
- record evidence of their progress, in an on-going critical and analytical review
- respond to a theme, stimulus or ideas
- make connections between their investigations and creative intentions
- realise intentions
- produce and present outcome(s).

Additional guidance

Drawing

Drawing is an essential skill for studying art and design at GCE and degree level. It forms a core element of the practice of artists, craftspeople and designers. It can take many forms; at its simplest and most direct it consists of marks of pencil or pen on paper, though it can employ any media and be applied in two and three dimensions or time-based media.

Drawing in the context of this qualification is taken to mean the following: recording the observed world, using mark-making in appropriate media; exploring ideas visually, through the act of mark-making; investigating, through the exploration of drawing media to find new ways of expressing ideas, feelings or observations; experimenting with various tools, materials and techniques. Further details are given in the following sections.

Annotation

Notes can be used to annotate visual work, to reflect on students' own work and that of others, as a development log and to record notes on experiments, ideas or visits. Students should aim to use specialist terminology and vocabulary relevant to art and design in their annotations.

Personal study

The personal study, which is an assessed element of Component 1, is a separate piece of writing which must comprise a minimum of 1000 words of continuous prose. Further details are outlined in the section on Component 1, pages 26–28.

Critical and contextual knowledge and understanding

Along with written annotation, an important aspect of student progression in art and design is the application of critical and contextual knowledge and understanding, and an ability to express and justify ideas. Developments of these aspects are integrated within practical work, written annotation and the spoken word. They may be honed, for example, through practical visual analysis; by using the spoken word in a critique, a discussion or an interview; through investigation of some aspect of art and design, e.g. from a museum, gallery or site visit.

Students should develop a critical understanding of their own work and appreciate it in relation to historical and contemporary contexts. In the broadest sense, this may be through looking at art, craft and design in galleries and museums, through various media, in the designed world and in understanding how art and design relates to everything around them.

Students should consider the impact of global influences, cultures and traditions both past and present, particularly those that are influential in relation to technique, colour, symbolism, ritual, social climate and location. Environmental and social issues have long been a source of work and debate for artists and designers, with issues such as poverty, sustainability, consumption and political movements stimulating points of creative investigation.

Where appropriate, students should make reference in their development work to influences from past and contemporary creative practice. They should become familiar with the work of artists, craftspeople and designers who have made a major contribution to practical and conceptual development in their area. They should be familiar with contemporary and emerging concepts and learn how to analyse and critically evaluate work, demonstrating an understanding of purposes, meanings and contexts.

Students should make links between investigation, drawing, design development, project management, technical and digital applications, final presentation and evaluation. There should be a continuous critical response to their own work, through methods such as an on-going learning log or blog, annotations and written explanations of technical and creative processes as well as notes about references and sources of information.

The critique is used in higher education as a vehicle for developing the student's ability to use specialist terminology and vocabulary, to listen to and respond to others' interpretations, to defend their own work and to express their thoughts and feelings on art and design. This can be an effective tool for students' to practice contextualising and verbalising an understanding of art and design concepts in relation to their own work.

Evidence of a student's critical and contextual knowledge and understanding will be found, for example, in annotated sketchbooks, visual studies, research notes into artists and designers, reviews and evaluations, and in creative expression of intentions.

Discrete evidence of in-depth study, and critical and contextual knowledge and understanding, will also be shown through the personal study of a minimum 1000 words of continuous prose, as outlined in Component 1, pages 26–28

Digital and non-digital media

While digital media offers new possibilities for making art and design, many of the underlying principles are common to non-digital media. Formal elements apply in the same way, although there are fundamental differences between screen-based and print-based images, such as the potential for different forms of interaction.

Many types of digital media are relatively new, while others extend the functions of non-digital media. They are subject to continual change in techniques and processes both in their own right and in the way they interact with non-digital media. Digital media can function as a tool, a process, a method of dissemination and a means of interaction with any audience.

Students can work entirely in digital media or entirely in non-digital media, or in any combination of the two, provided the Assessment Objectives are addressed.

Further guidance on how digital and non-digital media may be applied is given in the title sections.

Depth of study

Through following this Advanced Level specification, it is expected that students will achieve in-depth knowledge, understanding and skills in art and design that will aid their progression to foundation and degree-related study in this area.

In-depth knowledge, understanding and skills can be achieved through:

- advanced specialisation in particular disciplines, techniques or processes
- extended development of particular themes, ideas or issues
- theoretical research and showing increased understanding
- integration of practical, written and other forms of communication
- rigorous exploration of an interdisciplinary or multidisciplinary approach.

Art, Craft and Design (9AD0)

Introduction

The broad-based Art, Craft and Design title will enable students to explore a range of two-dimensional, three-dimensional or time-based approaches to their studies.

Artists, craftspeople and designers, while working within their own disciplines, are often inspired and influenced by those working in a wide range of different creative areas. They draw inspiration from images, compositions, forms and ideas outside of their own discipline. They work frequently in multi-disciplinary and cross-disciplinary ways, breaking traditional boundaries to create works of art, craft and design that combine ideas, concepts and materials in new ways.

Artists, craftspeople and designers all follow a creative process, in which they explore and develop approaches to communicating ideas visually. When generating and developing new ideas they are pro-active, persistent and self-motivated. Producing creative, dynamic and exciting outcomes, they can inspire and challenge the viewer through interaction with their work and engagement with their ideas.

Disciplines within Art, Craft and Design

Students should be encouraged to work and develop skills in a multi-disciplinary and cross-disciplinary way, exploring the connections between areas of art, craft and design and gaining knowledge and understanding of the scope and variety of disciplines and approaches.

Students undertaking the Art, Craft and Design title will work in disciplines chosen from **at least two** of the other five endorsed titles for Component 1, with the option to specialise in Component 2 by producing work in **one or more** disciplines. The disciplines available are: painting and drawing, printmaking, sculpture, lens-based image making, advertising, illustration, branding, information design, textiles for interiors, fine art textiles, fashion textiles, spatial design, product design, design crafts, film-based photography, digital photography, and film and video.

For Component 1: Personal Investigation, students must work in **at least two** disciplines, drawn from different endorsed titles.

For Component 2: Externally Set Assignment, students may choose to produce work in **one or more** disciplines.

Students may produce work entirely from a single discipline from one of the endorsed titles for Component 2. If they do so, their work will be standardised with the work from students taking that endorsed title.

Please refer to the endorsed title sections for further details.

Art and Design (Fine Art) (9FA0)

Introduction

Fine art requires engagement with aesthetic and intellectual concepts through the use of traditional and/or digital media, materials, techniques and processes for the purpose of self-expression, free of external constraints. Fine art may be created to communicate ideas and messages about the observed world, the qualities of materials, perceptions, or preconceptions. It can also be used to explore personal and cultural identity, society and how we live, visual language, and technology. Fine Art allows us to consider and reflect on our place in the world, both as individuals and collectively.

Drawing and other materials processes

Drawing in fine art forms an essential part of the development process from initial idea to finished work; from rough sketches, to diagrams setting out compositions, to digital drawings used for installations or as part of three-dimensional work. Students should use a variety of tools, materials and techniques, as appropriate, for recording their surroundings and source materials. Students should consider the application and implications of new and emerging technologies that can be used in conjunction with traditional and digital fine art materials.

Contextual understanding and professional practice

Contexts for fine art can be found in a wide range of sources; for example, from historical works in museums, contemporary art shows and fairs, an exhibition at a local gallery, films, architecture, music, literature and nature.

Students are required to develop the knowledge, skills and understanding outlined on pages 5 and 6. When undertaking work in fine art, students should also engage with:

- concepts such as figuration, representation and abstraction
- how the formal elements evoke responses in the viewer
- various forms or presentation in fine art and the ways that audiences may respond to or interact with them
- sustainable materials and production processes in the construction of work
- the potential of collaborative working methodologies in the creative process

Disciplines within fine art

For the purposes of this qualification, fine art is sub-divided into the following four disciplines:

- painting and drawing
- printmaking
- sculpture
- lens-based image making.

Students will be required to work in one or more of the disciplines to communicate their ideas. By working across disciplines, they will extend their understanding of the scope of fine art; by focusing on one discipline, they will gain a deeper understanding of specific processes within fine art.

Painting and drawing

Students will develop integrated knowledge, skills and understanding of the following:

- characteristics of materials such as plasticity, opacity, translucence, malleability and transparency
- properties of colour, such as hue, tint, saturation, tone and colour perception
- materials such as graphite, wax crayon, oil pastel, soft pastel, aquarelle, charcoal, ink, chalk, conté crayon, gouache, watercolour, acrylic paint, oil paint, dyes and computer software
- the use of a range of tools, such as artists' brushes, decorators' tools, knives, sponges, digital software, fingers, card squeegees, scrapers, sticks, found objects and natural forms
- the potential for exploring combinations of materials, such as combining drawn and painted elements, collage, found objects, including inert materials to add textures/impart meaning.

Printmaking

Students will develop integrated knowledge, skills and understanding of the following:

- print qualities and how they result from different printmaking tools, materials and processes
- printing processes such as screen printing, intaglio printing, relief printing.

Sculpture

Students will develop integrated knowledge, skills and understanding of the following:

- producing forms in three dimensions, utilising volume, space, materials and movement

- carving techniques such as cutting and abrading
- construction techniques such as fixing or joining materials such as card, metals and plastics, using processes such as soldering, brazing, welding, glueing, jointing, riveting, bolting and 3D printing
- materials such as wood, stone, plaster, leather, clay, textiles, card, plastics, wax, recyclable materials, ready-mades and found objects/materials.

Lens-based image making

Students will develop integrated knowledge, skills and understanding of the following:

- the production processes of artworks in a range of lens and time-based media, such as mixed media, installation, site-specific, montage, digital, film and video, animation and sound
- elements that can contribute to lens-based image making such as lighting, sets, environments and sound
- qualities and functions of various film and video formats, such as 8mm film, analogue video, digital video, HD, 4K, .flv, .mov, .wmv, animated gif
- editing, including knowledge of the variety of ways in which images might be juxtaposed to create appropriate effects, such as in-camera editing, non-linear, offline edits, use of timekey, compression, in and out points.

Art and Design (Graphic Communication)

(9GC0)

Introduction

Graphic communication conveys information and ideas through visual means. The critical element for a graphic designer is the effective communication of a message or idea through the organisation of images and words. The scope of graphic communication has been extended through the growth of design applications in the home and in public and through the development of the internet. Graphic designers need to understand user and audience needs and how these groups respond to various forms of visual communication; as well as how changes to working practices lead to new forms of communication and presentation.

Drawing and other materials processes

Drawing in the context of graphic communication forms an essential part of the development process from initial idea to finished product; from rough sketches to diagrams setting out designs, including digital drawings. Students should use a variety of tools and materials, as appropriate, for recording their surroundings and source materials. Students must be alert to the possibilities offered by a range of materials, techniques and processes within graphic communication and of the important role of signs and symbols. They must be able to balance aesthetic and commercial considerations when producing graphic solutions to defined problems.

As well as developing skills in their use, students should become aware of and be able to judge when it is appropriate to use traditional or computer-based methods within graphic communication processes, for example in developing roughs, layouts and mock-ups of potential design solutions, in recording and developing ideas and for final designs and presentation. Sometimes ideas or feelings need to be recorded and developed rapidly; the keys to rapid execution are familiarity with and availability of materials. Students should consider the application and implications of new and emerging technologies that can be used in conjunction with traditional and digital graphic communication materials.

Contextual understanding and professional practice

Contexts for graphic communication can be found in a wide range of sources; for example, from historical collections and museums, contemporary graphic communication shows and fairs, the local environment of signage, advertising and branding, films, architecture, music, literature and nature.

Students should consider the issues, opportunities and constraints involved in image and content copyright. They should be aware of the circumstances and conditions in which it is acceptable to incorporate images and design concepts originated by others, and of the appropriate steps to take to ensure permission to reproduce their own work is suitably managed.

Students are required to develop the knowledge, skills and understanding outlined on pages 5 and 6. When undertaking work in graphic communication, students should also engage with:

- how audiences may respond to the use of words, images and how the formal elements evoke responses in the viewer
- the basic typographical and layout requirements for digital and print-based products
- sustainable materials and production processes in the construction of work
- the potential of collaborative working methodologies in the creative process.

Disciplines within graphic communication

For the purposes of this qualification, graphic communication is sub-divided into the following four disciplines:

- advertising
- illustration
- branding
- information design.

Students will be required to work in one or more of the disciplines to communicate their ideas. By working across disciplines, they will extend their understanding of the scope of graphic communication; by focusing on one discipline, they will gain a deeper understanding of specific processes within graphic communication.

Advertising

Students will develop integrated knowledge, skills and understanding of the following:

- how graphic communication is used to convey information, arouse interest, tell stories, create brand recognition, sell a product or service, promote brand loyalty
- the role of graphic communication within marketing strategies, promotional campaigns, corporate identity design, logo design
- design briefs, clients, audiences, web-based and digital advertising, use of social media
- the use of images and typography in advertising, such as photography, animation and video.

Illustration

Students will develop integrated knowledge, skills and understanding of the following:

- the relationships between illustration and narrative
- illustration briefs, clients and audiences

- illustration for a variety of purposes, such as book, magazine, advertising, covers, web-based, interactive
- infographics as a way of communicating data through imaginative charts and diagrams.

Branding

Students will develop integrated knowledge, skills and understanding of the following:

- how packaging is determined by its contents
- marketing briefs, clients and audiences, brand identity, brand loyalty
- making suitable production drawings, which may include computer-generated ideas and developments
- surface images, illustration decoration or pattern for packaging
- development and construction of three-dimensional prototypes, considering production materials, recyclable design, using sustainable or renewable materials
- planning and developing procedures for reproduction and manufacture
- specifying sustainable materials and production processes that are suitable for recycling and/or reuse
- the legal requirements for information that must be included on certain types of packaging, and for barcoding and tracking.

Information design

Students will develop integrated knowledge, skills and understanding of the following:

- letter forms, font types, serif and sans serif fonts, leading, paragraph indents, hanging indents, justification, alignment, headings, kerning and sub-headings
- typographical requirements for digital and print-based products, such as magazine design, newspaper design, web page design, leaflet and poster design
- the appropriate use of templates, page layout, style sheets, image manipulation, compression, workflow and file types
- 3d digital graphic techniques, such as modelling objects, rotation, lathing, extruding, linking, creating and applying textures and lighting effects
- moving image/time-based digital graphic techniques, such as storyboarding, sound, animation, colour consistency
- the appropriate use and combination of words, visual 2D and 3D representations, time and desired user behaviours and responses
- interface design, such as the use of symbols and words to aid and enhance navigation, principles of control panel display, use of navigational structures and levels

Art and Design (Textile Design) (9TE0)

Introduction

Textile design is a versatile discipline that involves the creation, selection, manipulation and application of a range of materials and processes in order to produce fabrics for different purposes. Textile designers are expected to work in a multi-disciplinary way to create ideas, concepts, materials and techniques for different applications; for example in fashion and clothing, there are specialised areas such as lingerie, nightwear, sportswear, accessories or innovative clothing for theatre or film production. Technological innovation in textile development is used increasingly to produce specialist fabrics, which can be protective, fireproof, waterproof, or performance enhancing.

Some textile designers create fabrics for interior applications such as upholstery, bed linens, curtains, wallpapers, carpets and rugs or work more generally as surface designers for vinyl coverings, tiles, leather, glass, packaging, greetings cards and wrapping paper. Textile design can also be interpreted as a personal and individual fine art discipline, being used to create installations, mixed media constructions, tapestries, three-dimensional pieces, applications, soft sculpture and wall hangings.

Drawing and other materials processes

At the core of any textile art and design related practice is drawing. Drawing is an essential skill in the development process, from initial visual research, recording from primary sources, through to the translation of ideas into finished designs. The practice of drawing in textile design embraces traditional and non-traditional tools with digital translations and applications, whether these relate to fine art based textiles, fashion and clothing, interiors or surface design.

Students should be aware of and apply traditional values of drawing, such as line, tone and shape in conjunction with digital applications for sewing, knitting, weaving and embroidery, as well as digital imaging software. Students could use drawing, photography and video as tools for recording their surroundings and source materials. A range of drawing methods will encourage students to consider multiple ways of reading, thinking and applying drawing in different contexts. This also enables textile design students to consider drawing as a tool of translation, analysis, design and illustration. Students should use a variety of tools and materials, as appropriate, for recording their surroundings and source materials.

Students should consider the application and implications of new and emerging technologies that can be used in conjunction with traditional and digital textile materials.

Contextual understanding and professional practice

Contexts for textile design can be found in a wide range of sources; for example, from galleries, historical collections and museums, contemporary trade fairs and fashion shows, a local exhibition, films, architecture, music, literature and nature.

Students should gain an awareness of environmental and global factors relating to textile design as well as a clear consideration of the aesthetics of the end product.

In addition, it is important that students are aware of current trends in textile and fashion design and development in order to achieve contextual currency in their work. This can be done through investigation into textile and fashion consultancies and agencies. Students should be familiar with contemporary and emerging concepts and learn how to analyse and critically evaluate textile design, demonstrating an understanding of purposes, meanings and contexts.

Students are required to develop the knowledge, skills and understanding outlined on pages 5 and 6. When undertaking work in textile design, students should also engage with:

- a range of materials and tools used for textiles processes, including ethically and ecologically considered sourcing
- how the formal elements evoke responses in the viewer
- the need for the specification of sustainable materials and energy-saving production processes in the construction of works and products
- sustainable materials and production processes in the construction of work
- the potential of collaborative working methodologies in the creative process.

Disciplines within textile design

For the purposes of this qualification, textile design is sub-divided into the following three disciplines:

- textiles for interiors
- fine art textiles
- fashion textiles.

Students will be required to work in one or more of the disciplines to communicate their ideas. By working across disciplines, they will extend their understanding of the scope of textile design; by focusing on one discipline, they will gain a deeper understanding of specific processes within textile design.

Textiles for interiors

Students will develop integrated knowledge, skills and understanding of the following:

- the appropriate and necessary skills, which may include computer-generated ideas for development in colour, materials and construction, design, repeat pattern, surfaces and materials
- knowledge and application of techniques such as weaving, knitting, embroidery, appliqué and fused or felted textiles
- the basis of textile or surface design printing, including croquis and repeating pattern
- techniques such as mono-print, transfer-relief, relief-print, wood-cut, lino-cut, silk-screen and digital printing

Fine art textiles

Students will develop integrated knowledge, skills and understanding of the following:

- a range of materials and tools for making fine art textiles, including, where appropriate, digital
- ways in which ideas and feelings can be effectively communicated through the organisation of materials and formal elements
- how audiences may respond to the variety of forms that work may take, including methods of presentation
- the appropriate skills to combine and exploit a range of media, materials and approaches that reflect contemporary fine art textile practice
- a working knowledge of a range of techniques.

Fashion textiles

Students will develop integrated knowledge, skills and understanding of the following:

- the creation and development of fashion design ideas, which may include drawing or recording skills, digital designs and the preparation of toiles or samples
- the appropriate skills to combine and explore a range of materials and approaches
- the appropriate skills of modelling, cutting, joining and embellishing in textiles or a range of appropriate media
- a working knowledge of a range of textile techniques chosen from other textile disciplines
- links to the broader areas of fashion, such as marketing, promotion and styling.

Art and Design (Three-dimensional Design) (9TDO)

Introduction

Three-dimensional design includes a range of art, craft and design practices to meet a specific need with sensitivity to both function and aesthetics. Form, function, materials and ultimate destination and use of the design or artefact are all important considerations for the three-dimensional designer. Students should be able to realise the full potential of their ideas and designs through the application of appropriate technical processes, though outcome(s) are likely to be in the form of prototypes, models and other forms of presentation.

Drawing and other materials processes

Drawing for designers forms an essential part of the development process from initial idea to finished product; from rough sketches to diagrams setting out patterns, to digital drawings used for products, spatial design or as an adjunct to design craft. Students should use a variety of tools and materials, as appropriate, for recording their surroundings and source materials.

Drawing is often an essential two-dimensional stage in the development of three-dimensional objects. Students will need to have an understanding of the way a three-dimensional world is depicted in two dimensions, for example by applying the rules of three-point perspective and the tools and conventions of 3D design software. Sometimes ideas or feelings need to be recorded and developed rapidly; the keys to rapid execution are familiarity with and availability of materials.

Students should be alert to the possibilities offered by a range of materials and the role of digital media in three-dimensional design. As well as developing skills in their use, students should become aware of and be able to judge when it is more appropriate to use traditional or computer-based methods of design and presentation. Students must continue to develop their understanding and application of the formal elements, and in particular the use of texture, colour and light, giving consideration to psychological and physical impact on the viewer in order to evoke a positive emotional response. Students should consider the application and implications of new and emerging technologies that can be used in conjunction with traditional and digital three-dimensional design materials.

Contextual understanding and professional practice

Contexts for three-dimensional design can be found in a wide range of sources; for example, from galleries, historical collections and museums, contemporary design and craft fairs, the built environment, restoration projects, products, craft items, films, architecture, music, literature and nature.

Students should understand the appropriate application of a range of design skills, including concept formulation and analysis of a brief, experimentation, evaluation and communication to a range of different audiences. They should investigate specific areas of a design problem, determine relevant sources of information and use them to research and explore possible solutions. Students should be familiar with contemporary and emerging concepts and learn how to analyse and critically evaluate three-dimensional design, demonstrating an understanding of purposes, meanings and contexts.

Students are required to develop the knowledge, skills and understanding outlined on pages 5 and 6. When undertaking work in three-dimensional design, students should also engage with:

- the properties and characteristics of a range of materials and tools used for modelling and manufacturing places, spaces and three-dimensional products including, where appropriate, digital
- the requirements for space, light, shape, form and colour in different design contexts and how they can be used to evoke emotional responses
- how the formal elements evoke responses in the viewer
- sustainable materials and production processes in the construction of work
- the potential of collaborative working methodologies in the creative process.

Disciplines within three-dimensional design

For the purposes of this qualification, three-dimensional design is sub-divided into the following three disciplines:

- spatial design
- product design
- design crafts.

Students will be required to work in one or more of the disciplines to communicate their ideas. By working across disciplines, they will extend their understanding of the scope of three-dimensional design; by focusing on one discipline, they will gain a deeper understanding of specific processes within three-dimensional design.

Spatial design

Students will develop integrated knowledge, skills and understanding of the following:

- the way in which spatial design incorporates the design of performance spaces, interiors, exhibitions, the built environment and architecture
- developing ideas involving scale drawings and models, plans, elevations, cross-sections and, where appropriate, perspectives, visualisations, models, fly-throughs and axonometric drawings, props, costumes and maquettes
- the demands of text, script, sound, choreography, stage directions, production, screenplay in relation to spatial design

- the role of technology in contemporary interiors and public spaces, including aspects such as service design, intelligent lighting, energy-saving devices and interactivity.

Product design

Students will develop integrated knowledge, skills and understanding of the following:

- a range of consumer product types, such as mechanical, electronic and decorative
- the generation and development of ideas, through appropriate drawing skills, computer-aided designs and the preparation of maquettes, prototypes and models
- real and virtual modelling, bending, cutting and fastening in materials such as clay, wood, plastic, card, glass, metal, textiles, including recycled materials
- influencing factors, such as interface and service design, portability, maintainability, miniaturisation and potential for shared information
- the realisation of product design concepts through an understanding of the properties and characteristics of a range of materials and processes
- product design as a component of branding, market position and opportunity and the ways in which customers relate to different products
- batch or mass production, core industrial methods and the potential impact of new production concepts.

Design crafts

Students will develop integrated knowledge, skills and understanding of the following:

- the properties and characteristics of a range of materials and processes, such as, clay, wood, plastic, card, glass, metal and textiles
- the potential of a craft-based approach in the production of artefacts such as jewellery, furniture, lighting, textiles, containers and toys
- the ways in which different materials can be combined in the production of artefacts
- the routes to market for craft-based items, such as specialist shops, tourist locations, events, exhibitions, online galleries
- the design of commissioned pieces for specific locations, individuals, clients or organisations.

Art and Design (Photography) (9PY0)

Introduction

Photography has been used by practitioners to record, document and present examples of everyday life, in ordinary and extraordinary circumstances. It has also been used as the vehicle for artistic expression, communicating personal ideas about the world around us. It is used to convey personal identity more widely than any other art form, is applied in the creative process across art, craft and design and is widely used in social, commercial and scientific contexts. The development of affordable lens-based technology has changed the way that both professionals and the public use photography.

Drawing and other materials processes

The word photography could be taken to mean 'a graphic representation with light'. In this way a photograph can take on the qualities of a drawing. In the context of this endorsed title, drawing forms an essential element of both development and final product. A camera can record the observed world but is not able on its own to explore ideas. Students must reflect on, refine and apply the observations they make with a camera, and determine which tools or techniques are most appropriate in their exploration of ideas. Drawing methods such as pen or pencil on paper may enhance their development and understanding of photographic ideas, for example to plan shots, analyse and deconstruct their own imagery, or record ways in which practitioners have used formal elements and visual language. Students should use a variety of tools and materials, as appropriate, for recording their surroundings and source materials.

Photography includes works in film, video, digital imaging and light-sensitive materials. Sometimes specific techniques and processes are used to convey messages and create works related to other disciplines, such as web-based animations, photographic images in printed journals, and light projections within theatrical or architectural spaces.

Many practitioners define their image before it has even been taken by scouting locations and by planning a shot around specific weather conditions or time of day, using filters, studio lighting, reflectors, soft boxes, props, makeup, or backgrounds to control each element within the frame.

Students should consider the application and implications of new and emerging technologies that can be used in conjunction with traditional and digital photography materials.

Contextual understanding and professional practice

Contexts for photography can be found in a wide range of sources; for example, from galleries and museums, contemporary photography shows, web-based sources, films, architecture, music, literature and nature.

Students must consider the issues, opportunities and constraints involved in image and content copyright. They should be aware of the circumstances and conditions in which it is acceptable to incorporate images and content originated by others, and of the appropriate steps to take to ensure permission to reproduce their own work is suitably managed. Students should be familiar with contemporary and emerging concepts and learn how to analyse and critically evaluate photography, demonstrating an understanding of purposes, meanings and contexts.

Students are required to develop the knowledge, skills and understanding outlined on pages 5 and 6. When undertaking work in photography, students should also engage with:

- the operations and principles of creating a photographic image, including the use of available and controlled light, lenses, cameras and light-sensitive materials, including digital and non-digital
- a range of materials used in photography, including print and screen-based materials
- how the formal elements evoke responses in the viewer
- the processes for production of digital and print-based photographs
- methods of presentation of photographic images
- sustainable materials and production processes in the construction of work
- the potential of collaborative working methodologies in the creative process.

Disciplines within photography

For the purposes of this qualification, photography is sub-divided into the following three disciplines:

- film-based photography
- digital photography
- film and video.

Students will be required to work in one or more of the disciplines to communicate their ideas. By working across disciplines, they will extend their understanding of the scope of photography; by focusing on one discipline, they will gain a deeper understanding of specific processes within photography.

Film-based photography

Students will develop integrated knowledge, skills and understanding of the following:

- film types, film speeds, specialised films which will facilitate the processes of generating and developing ideas, pushing/pulling films, reciprocity failure
- viewpoint, composition, focus, aperture, shutter speed, exposure, through the lens metering
- darkroom techniques, using appropriate paper types, developing and printing, emulsions, exposures, tone and contrast

- acquisition, manipulation and distribution of the image through computers, scanners, photocopiers and computer software.

Digital photography

Students will develop integrated knowledge, skills and understanding of the following:

- the principles of digital photography, including the pixel and digital processing
- viewpoint, white balance, composition, focus, aperture, shutter speed, exposure, shooting modes, histograms
- the use and qualities of image acquisition hardware and software, image manipulation and analogies between digital and other forms of photography
- the relationships between colour and tone for screen and print-based media, screen calibration, colour gamut, file formats such as raw, jpeg, tiff, png
- the use of a range of source material, software and hardware in the generation and development of ideas.

Film and video

Students will develop integrated knowledge, skills and understanding of the following:

- synopsis, storyboards, scripting, camera angles, viewpoints, length of shot, cutting, composition, cropping and pacing, which may include computer-generated ideas and developments
- various animation processes, such as stop-frame, rostrum and 3D modelling and associated hardware and software
- qualities and functions of various film and video formats, such as 8mm film, analogue video, digital video, HD, 4K, .flv, .mov, .wmv, animated gif
- the use of sound, narration and storyline and their relation to moving images
- editing, including knowledge of the variety of ways in which images might be juxtaposed to create appropriate effects, such as in-camera editing, non-linear, offline edits, use of timekey, compression, in and out points.